

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)

Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total Marks: 50
Q.N o	Question	Bloom's Level	Marks
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	5
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	5
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	5
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	5
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	5
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	5
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	5

8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	5
9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	5
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	5

Code of Conduct:

I _ N i s h a . R (1 9 2 3 1 1 4 1 5) ____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Nisha R

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated

Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

ANSWERS:

Q1. Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?

Answer:

Query processing converts an SQL query into an executable plan.

1. **Parsing:** Checks SQL syntax and semantics.
2. **Translation:** Converts SQL into relational algebra/query tree.
3. **Optimization:** Generates different execution plans.
4. **Plan Selection:** Compares plans using I/O, CPU, memory, indexes, and join costs.
5. **Execution:** Executes the selected plan and returns the result.

Conclusion: The optimizer chooses the plan with the lowest estimated cost for efficient execution.

Q2. What is the difference between heuristic optimization and cost-based optimization?

When would you use each?

Answer:

Heuristic optimization uses predefined rules such as pushing selections and projections down and joining smaller relations first.

Cost-based optimization estimates the cost of different plans using factors such as I/O, CPU, memory, indexes, and table size.

- **Heuristic:** Faster and suitable for simple queries.
- **Cost-based:** More accurate and suitable for complex queries.

Conclusion: Heuristic optimization is rule-based, while cost-based optimization selects the lowest-cost plan.

Q3. How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.

Answer:

In a Java project, database connectivity can be implemented using **JDBC**.

Steps:

1. Load the JDBC driver.
2. Create a database connection.
3. Create a Statement or PreparedStatement.
4. Execute the query.
5. Process the result.
6. Close the resources.

JDBC vs ODBC:

- JDBC is designed mainly for **Java applications**.
- ODBC is a more general database connectivity interface.
- JDBC uses JDBC drivers, while ODBC uses ODBC drivers.

Conclusion: JDBC is commonly preferred for Java-based database applications.

Q4. What are ACID properties? Give a real-world example where each property is critical.

Answer:

ACID properties ensure reliable database transactions.

1. **Atomicity:** A transaction is completely executed or completely rolled back.

Example: Bank transfer.

2. **Consistency:** Database rules must remain valid after a transaction.

Example: Account balance constraints.

3. **Isolation:** Concurrent transactions should not interfere incorrectly.

Example: Online seat booking.

4. **Durability:** Committed data remains safe after a failure.

Example: Completed online payment.

Conclusion: ACID properties maintain database reliability and consistency.

Q5. Explain concurrency control problems: dirty read, lost update, and unrepeatable read.

How does 2PL solve them?

Answer:

Common concurrency problems are:

- **Dirty Read:** Reading data written by an uncommitted transaction.
- **Lost Update:** One transaction overwrites another transaction's update.
- **Unrepeatable Read:** The same read gives different values because another transaction modified the data.

2PL has two phases:

- **Growing phase:** Locks are acquired.
- **Shrinking phase:** Locks are released.

Strict 2PL holds exclusive locks until commit, preventing dirty reads and cascading problems.

Conclusion: 2PL provides conflict serializability and improves concurrency control.

Q6. What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.

Answer:

A **deadlock** occurs when transactions wait for each other indefinitely.

Example:

T1 holds A → waits for B

T2 holds B → waits for A

Detection:

1. Create a **Wait-For Graph**.
2. Add an edge when one transaction waits for another.
3. Check the graph for a cycle.
4. A cycle indicates deadlock.
5. Roll back one transaction.

Prevention:

- **Wait-Die:** Older transaction waits; younger transaction is rolled back.
- **Wound-Wait:** Older transaction rolls back the younger transaction; younger transaction waits.

Conclusion: Deadlocks can be detected using cycles and prevented using timestamp-based methods.

Q7. Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?

Answer:

Immediate Update

Database may be updated before commit

Uses UNDO and REDO

Uncommitted changes may reach disk

More flexible

Deferred Update

Database updated after commit

Mainly uses REDO

Uncommitted changes are not written

Simpler recovery

Immediate Update: Suitable for systems requiring frequent and fast updates.

Deferred Update: Suitable when simpler recovery is preferred.

Conclusion: Immediate Update provides flexibility, while Deferred Update provides simpler recovery.

Q8. Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?

Answer:

Shadow Paging maintains two page tables:

- **Shadow page table:** Stores the stable database state.
- **Current page table:** Points to modified pages.

When a page is modified, a new copy is created. On **commit**, the current table becomes permanent. On **failure**, the current table is discarded and the shadow table is used.

Advantages:

- Simple recovery.
- Fast rollback.
- No traditional UNDO logging.

Limitations:

- Requires extra storage.
- Page copying can be expensive.
- Can cause fragmentation.

Conclusion: Shadow Paging is simple but less suitable for very large, highly concurrent systems.

Q9. What is a serializability schedule? How do you test for conflict serializability using a precedence graph?

Answer:

A **serializable schedule** produces the same result as some serial execution of transactions.

Precedence Graph:

1. Create a node for every transaction.
2. Identify conflicting operations.
3. If T_i executes before T_j for a conflict, add:

$T_i \rightarrow T_j$

4. Check for cycles.

- **No cycle → Conflict serializable**
- **Cycle → Not conflict serializable**

Conclusion: An acyclic precedence graph indicates conflict serializability.

Q10. Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?

Answer:

2PL has two phases:

- **Growing:** Acquire locks.
- **Shrinking:** Release locks.

Basic 2PL

Locks can be acquired during the growing phase and released during the shrinking phase. It guarantees conflict serializability.

Strict 2PL

Exclusive locks are held until commit or abort. It prevents dirty reads and cascading rollback.

Rigorous 2PL

Both shared and exclusive locks are held until commit or abort.

Most commonly used: Strict 2PL, because it provides good concurrency, consistency, and easier recovery.

Conclusion: Strict 2PL provides a good balance between performance and correctness.
